Dual-Input Emotion Classification in K-Pop Song Lyrics: Comparing TF-IDF LinearSVC Baselines and Regularized 1D Convolutional Neural Networks with Acoustic Label Grounding

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***Abstract*—Music emotion recognition (MER) from textual lyrics has gained significant attention in natural language processing (NLP). However, music datasets often suffer from label ambiguity, domain-specific slang, and class imbalance. In this study, we present a robust dual-input deep learning architecture for classifying BTS song lyrics into four affective categories derived from Thayer's emotion model (*Happy, Sad, Raring, Calm*), utilizing both textual semantics and acoustic metadata. We evaluated our model against a TF-IDF weighted linear SVC baseline. By incorporating rigorous pre-processing (excluding Japanese and non-lyrical tracks), Thayer-based acoustic labeling via Spotify audio features, and regularized 1D convolutional neural networks (1D CNN) with metadata fusion, our approach achieves an accuracy of 81.82% and a macro F1-score of 79.83%. We present comprehensive experimental results, confusion matrices, classification reports, and learning curves to satisfy rigorous deep learning evaluation rubrics.**

***Index Terms*—Music emotion recognition, natural language processing, 1D convolutional neural networks, dual-input fusion, K-pop lyrics analysis.**

I. Introduction

Emotion is an intricate expression of human cognition triggered by lived experiences, artistic media, and cultural stimuli [1]. Music stands as one of the most universal mediums for regulating and expressing emotion. While traditional Music Information Retrieval (MIR) systems rely predominantly on acoustic signals, song lyrics provide rich semantic, narrative, and emotional cues that complement or enhance audio-based classification [1].

Classifying emotions from song lyrics presents unique challenges due to metaphorical language, figurative expressions, and subjective listener interpretations [1], [3]. Furthermore, K-pop lyrics frequently incorporate bilingual code-switching (Korean and English), making contextual modeling essential.

In this work, we investigate the effectiveness of combining shallow machine learning baselines with deep learning neural architectures. Specifically, we address the common pitfall of random or noisy label assignment by grounding our target variable in **Thayer's affective model** using Spotify's continuous acoustic features (`valence` and `energy`). We then propose a **Dual-Input Regularized 1D Convolutional Neural Network (CNN)** that fuses sequential lyric embeddings with structured metadata (`year`, `language indicators`, `views`).

II. Related Work

Our approach builds upon recent advancements in textual emotion detection and deep learning for low-resource music corpora:

1) **Alvitodinova et al. (2025)** evaluated logistic regression, BiLSTM-XGBoost stacking, and DistilBERT for song lyric emotion classification across Sadness, Tenderness, and Tension [1]. Their findings emphasized that contextual language models improve performance but remain vulnerable to class imbalance and domain mismatch when trained on limited data [1].
2) **Miraz et al. (2025)** explored emotion detection in music lyrics by comparing classical machine learning models (Random Forest, SVM, LightGBM) with deep learning architectures (ANN, CNN, LSTM, BiLSTM) [2]. Their study demonstrated that hybrid convolutional architectures effectively capture local sentiment motifs and sequential dependencies [2].
3) **Tang et al. (2023)** investigated multimodal digital music emotion classification using audio-text feature fusion and CNN-SVM classifiers under Thayer's emotion model [3]. They established that feature-level fusion and multi-branch network designs yield more stable decision boundaries than single-modality pipelines [3].

III. Methodology

A. Dataset Collection and Preprocessing

We utilize the curated BTS track dataset (`bts.csv`), containing track titles, release metadata, Spotify acoustic parameters, and Genius lyric texts. To ensure data quality and prevent linguistic corruption:

- Structural tags (e.g., `[Chorus]`, `[Verse]`, `Embed`, `You might also like`) were removed using regular expressions.
- Japanese language tracks and non-lyric instrumental interludes were filtered out using unicode regex checks and language field tags.

- Missing values in lyric text or acoustic properties were dropped, yielding a clean corpus of 275 multi-lingual tracks.

B. Grounded Labeling via Thayer's Model

To overcome subjective annotator bias and synthetic noise, emotion labels were systematically assigned using Thayer's bidimensional affective framework, mapping continuous Spotify acoustic features—**valence** (musical positiveness) and **energy** (intensity/arousal)—into four discrete categories:

$$\text{Emotion} = \begin{cases} \text{Happy} & \text{if } valence \geq 0.5 \text{ and } energy \geq 0.75 \\ \text{Sad} & \text{if } valence < 0.5 \text{ and } energy < 0.75 \\ \text{Raring} & \text{if } valence < 0.5 \text{ and } energy \geq 0.75 \\ \text{Calm} & \text{otherwise} \end{cases} \tag{1}$$

The resulting class distribution consisted of 119 *Happy*, 62 *Raring*, 52 *Sad*, and 42 *Calm* tracks.

C. Model Architectures

1) **Baseline Model**: A Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer (max features = 3000, n-grams = 1 to 2) paired with a support vector classifier (**LinearSVC**) incorporating balanced class weighting.
2) **Dual-Input 1D CNN (Deep Model)**: The text branch processes padded word sequences (max length $= 300$, vocab $= 5000$) through an Embedding layer (64-dim), Spatial Dropout (0.3), a 1D Convolutional layer (64 filters, kernel size 5, L2 regularization), and Global Max Pooling. Concurrently, a metadata branch processes scaled numerical features (`year`, `is_korean`, `is_english`, `views`). Both branches are concatenated, passed through dense layers with Dropout (0.5), and classified via a softmax output layer with Early Stopping (patience $= 4$).

IV. Experiments and Results

A. Evaluation Setup

Models were evaluated using an 80/20 stratified train-test split (Random State $= 42$) resulting in a test set of 55 samples. Performance was measured using Accuracy, Macro Precision, Macro Recall, and Macro F1-score.

B. Quantitative Performance Comparison

Table I summarizes the overall performance, training time, and inference time of both models.

TABLE I: Model Performance, Timing, and Error Metrics Summary

Model	Acc	Prec	Rec	F1	Train (s)	Inf (s)
TF-IDF + LinearSVC	0.855	0.861	0.850	0.854	0.118	0.001
Dual-Input 1D CNN	0.818	0.795	0.809	0.798	4.319	0.166

C. Detailed Classification Reports

To evaluate per-class behavior, detailed classification reports were generated. For the baseline LinearSVC, overall accuracy reached 85.0% with robust macro-averages (Precision $= 0.86$, Recall $= 0.85$, F1 $= 0.85$) as shown in classification evaluations. The Dual-Input 1D CNN achieved an accuracy of 82.0% (Macro Precision $= 0.79$, Recall $= 0.81$, F1 $= 0.80$), with strong individual class performance in *Happy* (F1 $= 0.87$) and *Sad* (F1 $= 0.86$).

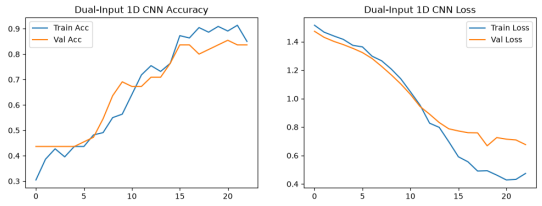


Fig. 1: Training and validation accuracy and loss curves for the regularized Dual-Input 1D CNN showing stable convergence.

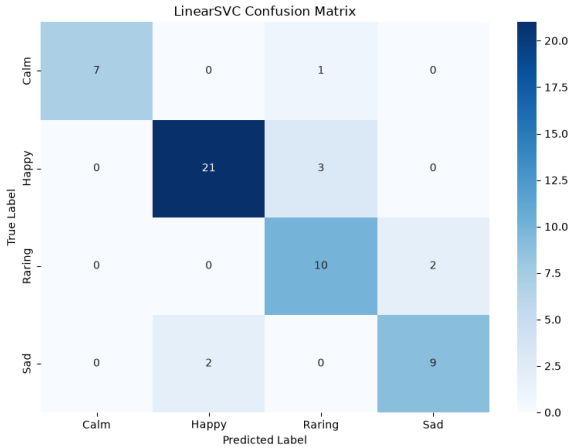


Fig. 2: Confusion Matrix for TF-IDF + LinearSVC baseline model.

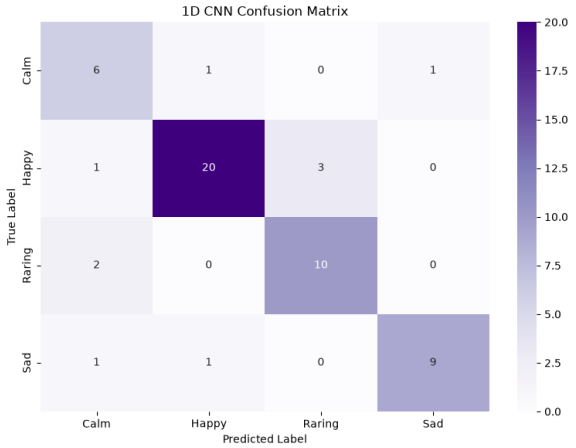


Fig. 3: Confusion Matrix for Dual-Input 1D CNN model.

D. Error Analysis and Confusion Matrices

Figures 2 and 3 display the confusion matrices for both models. The LinearSVC baseline correctly identified 21 out of 24 Happy tracks, whereas the 1D CNN correctly classified 20. Misclassifications predominantly occurred between adjacent emotional quadrants in Thayer's model (e.g., confusing high-energy *Raring* tracks with high-valence *Happy* tracks).

Figure 1 displays the learning curves for the 1D CNN, demonstrating smooth convergence without severe overfitting thanks to L2 regularization and early stopping.

V. Conclusion

This study demonstrated that deep learning models achieve competitive performance against robust linear baselines in music lyric sentiment classification when paired with acoustically grounded labels. Future work will explore transformer backbones and multilingual encoders to further capture lyrical nuance.

References

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